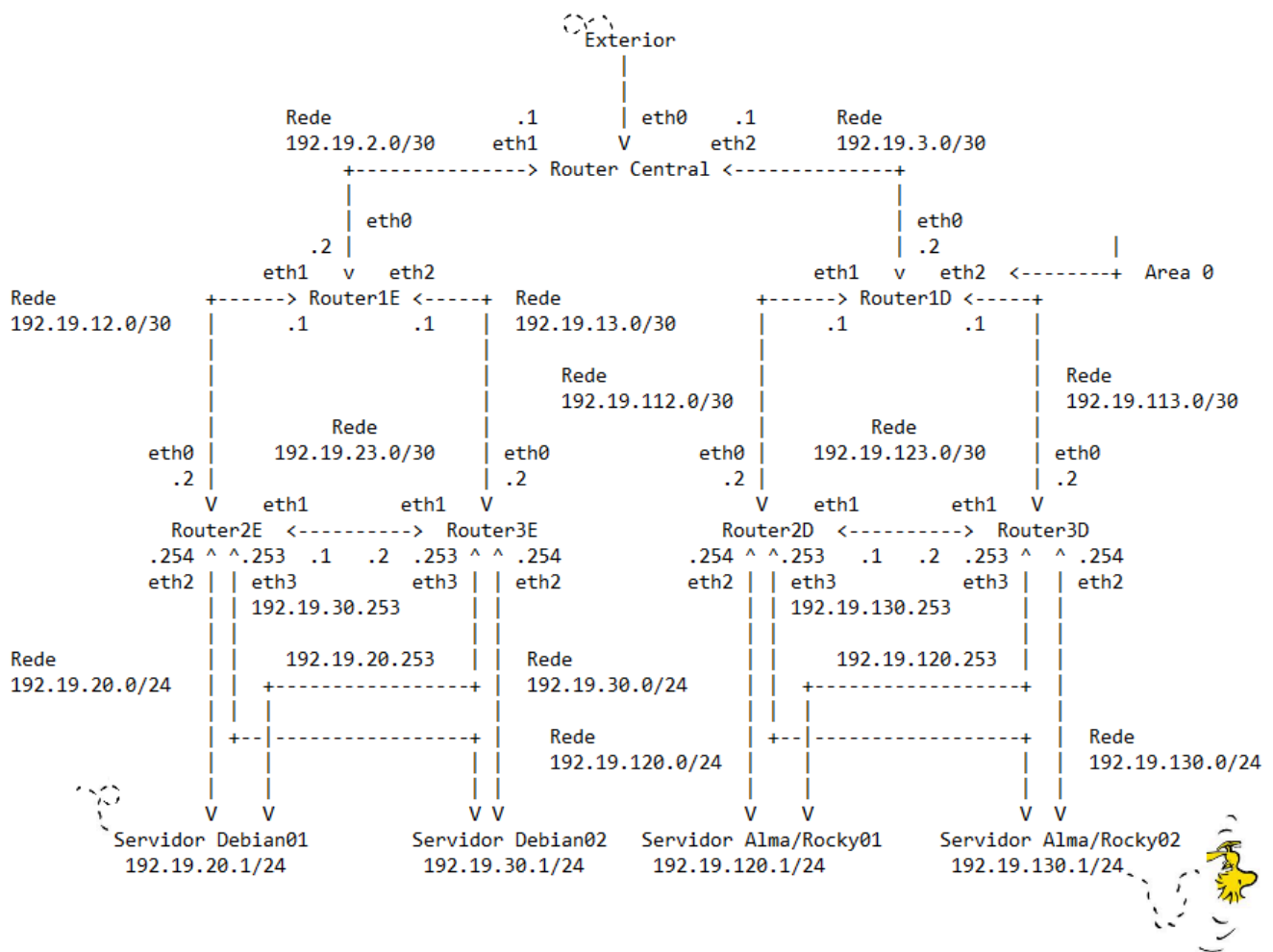


Simulação de Rede Empresarial

VyOS, OSPF, VPN



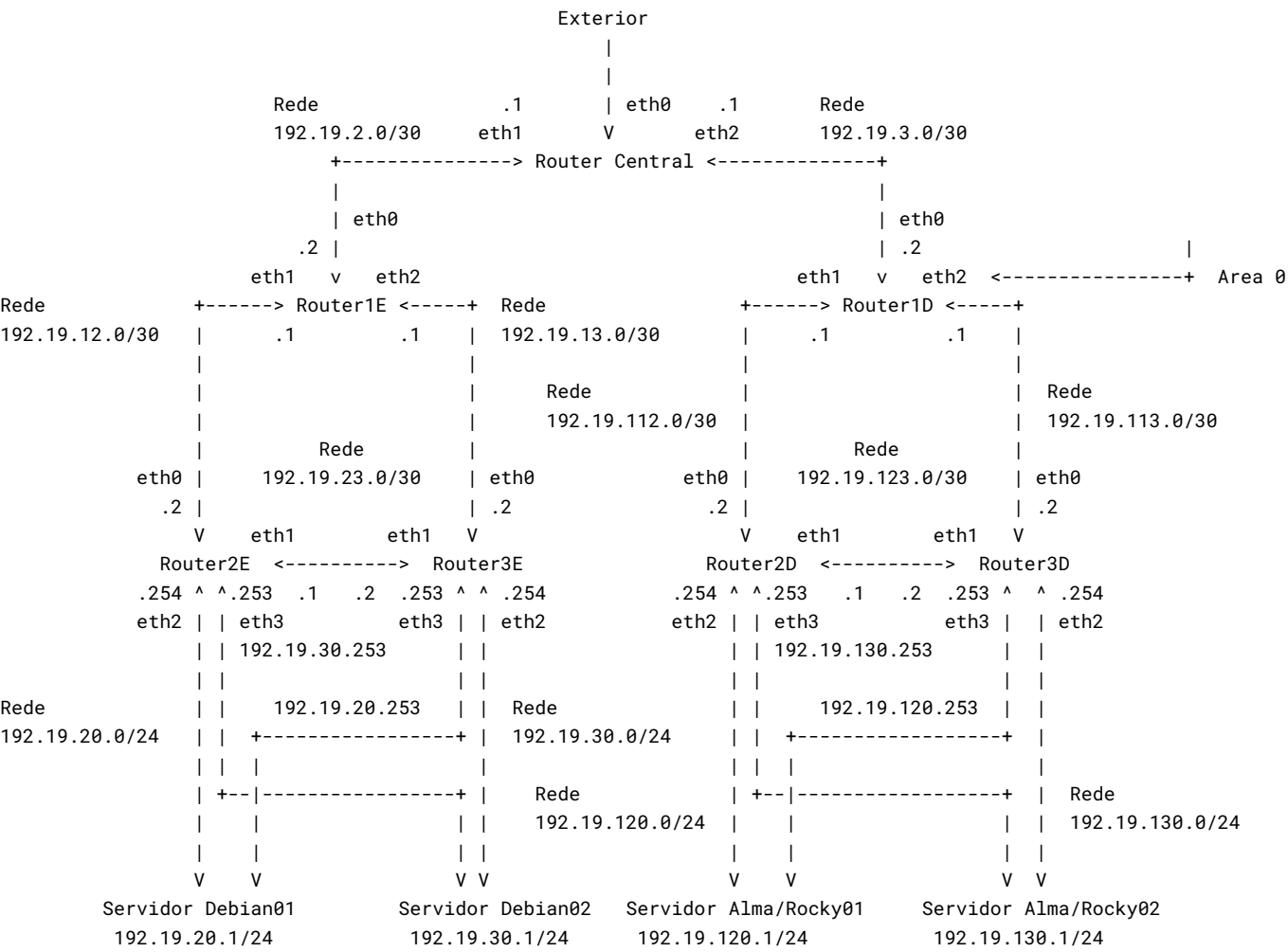
Pedro Nuno Vieira

2025-07-13

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Esquema de Rede



OSPF

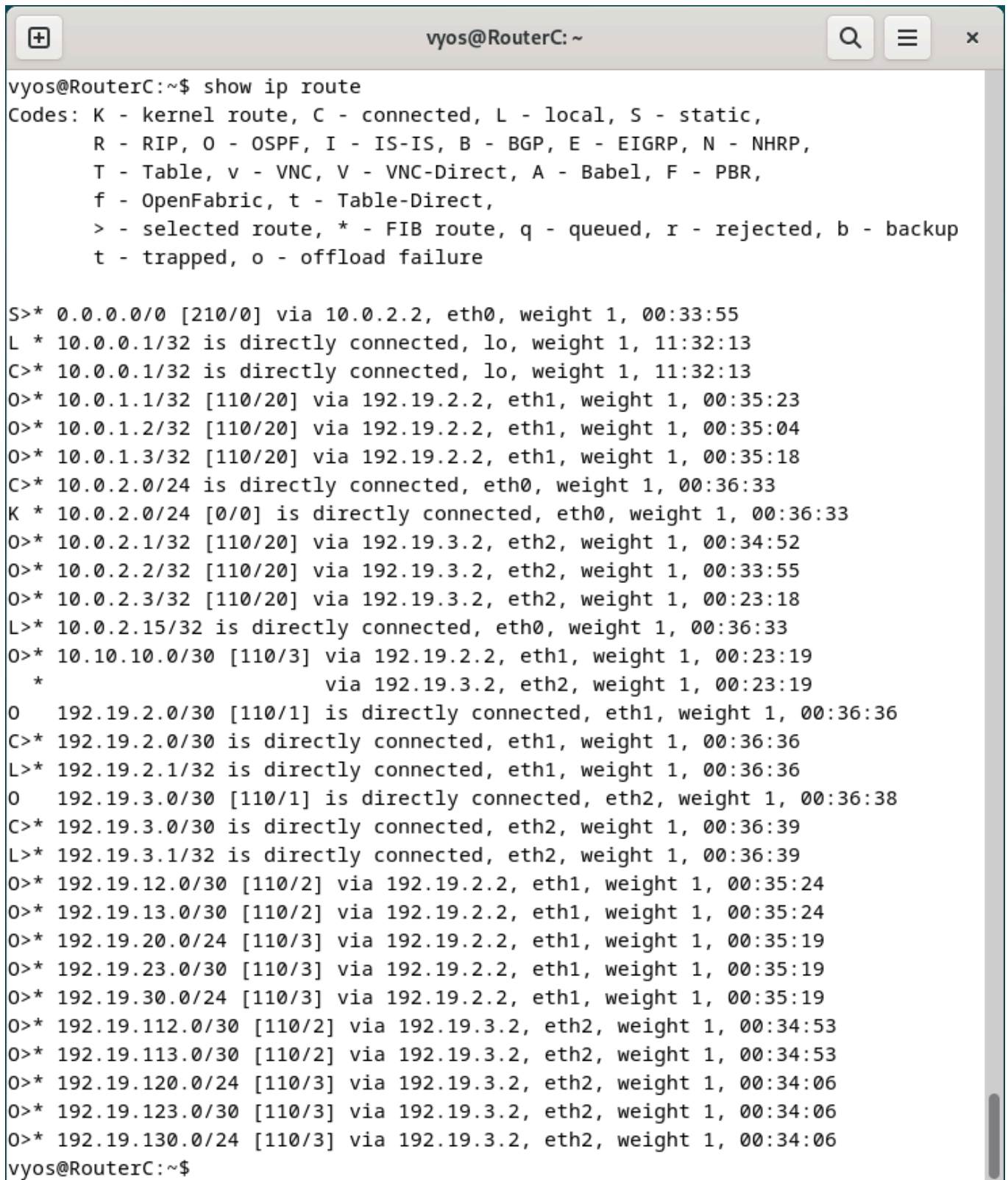
Area 0
RouterC
Router1E Router 1D

Area 1
Router1E
Router2E Router 3E

Area 2
Router1D
Router2D Router 3D

A. Prints de Routing/Encaminhamento - OSPF

Para cada router, mostrar `show ip route`



```
vyos@RouterC: ~  
vyos@RouterC:~$ show ip route  
Codes: K - kernel route, C - connected, L - local, S - static,  
        R - RIP, O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,  
        T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,  
        f - OpenFabric, t - Table-Direct,  
        > - selected route, * - FIB route, q - queued, r - rejected, b - backup  
        t - trapped, o - offload failure  
  
S>* 0.0.0.0/0 [210/0] via 10.0.2.2, eth0, weight 1, 00:33:55  
L * 10.0.0.1/32 is directly connected, lo, weight 1, 11:32:13  
C>* 10.0.0.1/32 is directly connected, lo, weight 1, 11:32:13  
O>* 10.0.1.1/32 [110/20] via 192.19.2.2, eth1, weight 1, 00:35:23  
O>* 10.0.1.2/32 [110/20] via 192.19.2.2, eth1, weight 1, 00:35:04  
O>* 10.0.1.3/32 [110/20] via 192.19.2.2, eth1, weight 1, 00:35:18  
C>* 10.0.2.0/24 is directly connected, eth0, weight 1, 00:36:33  
K * 10.0.2.0/24 [0/0] is directly connected, eth0, weight 1, 00:36:33  
O>* 10.0.2.1/32 [110/20] via 192.19.3.2, eth2, weight 1, 00:34:52  
O>* 10.0.2.2/32 [110/20] via 192.19.3.2, eth2, weight 1, 00:33:55  
O>* 10.0.2.3/32 [110/20] via 192.19.3.2, eth2, weight 1, 00:23:18  
L>* 10.0.2.15/32 is directly connected, eth0, weight 1, 00:36:33  
O>* 10.10.10.0/30 [110/3] via 192.19.2.2, eth1, weight 1, 00:23:19  
    *  
        via 192.19.3.2, eth2, weight 1, 00:23:19  
O 192.19.2.0/30 [110/1] is directly connected, eth1, weight 1, 00:36:36  
C>* 192.19.2.0/30 is directly connected, eth1, weight 1, 00:36:36  
L>* 192.19.2.1/32 is directly connected, eth1, weight 1, 00:36:36  
O 192.19.3.0/30 [110/1] is directly connected, eth2, weight 1, 00:36:38  
C>* 192.19.3.0/30 is directly connected, eth2, weight 1, 00:36:39  
L>* 192.19.3.1/32 is directly connected, eth2, weight 1, 00:36:39  
O>* 192.19.12.0/30 [110/2] via 192.19.2.2, eth1, weight 1, 00:35:24  
O>* 192.19.13.0/30 [110/2] via 192.19.2.2, eth1, weight 1, 00:35:24  
O>* 192.19.20.0/24 [110/3] via 192.19.2.2, eth1, weight 1, 00:35:19  
O>* 192.19.23.0/30 [110/3] via 192.19.2.2, eth1, weight 1, 00:35:19  
O>* 192.19.30.0/24 [110/3] via 192.19.2.2, eth1, weight 1, 00:35:19  
O>* 192.19.112.0/30 [110/2] via 192.19.3.2, eth2, weight 1, 00:34:53  
O>* 192.19.113.0/30 [110/2] via 192.19.3.2, eth2, weight 1, 00:34:53  
O>* 192.19.120.0/24 [110/3] via 192.19.3.2, eth2, weight 1, 00:34:06  
O>* 192.19.123.0/30 [110/3] via 192.19.3.2, eth2, weight 1, 00:34:06  
O>* 192.19.130.0/24 [110/3] via 192.19.3.2, eth2, weight 1, 00:34:06  
vyos@RouterC:~$
```

RouterC show ip route

```
vyos@Router1E: ~  
vyos@Router1E:~$ show ip route  
Codes: K - kernel route, C - connected, L - local, S - static,  
R - RIP, O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,  
T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,  
f - OpenFabric, t - Table-Direct,  
> - selected route, * - FIB route, q - queued, r - rejected, b - backup  
t - trapped, o - offload failure  
  
O>* 0.0.0.0/0 [110/10] via 192.19.2.1, eth0, weight 1, 00:41:20  
O>* 10.0.0.1/32 [110/20] via 192.19.2.1, eth0, weight 1, 00:41:20  
L * 10.0.1.1/32 is directly connected, lo, weight 1, 11:32:59  
C>* 10.0.1.1/32 is directly connected, lo, weight 1, 11:32:59  
O>* 10.0.1.2/32 [110/20] via 192.19.12.2, eth1, weight 1, 00:41:02  
O>* 10.0.1.3/32 [110/20] via 192.19.13.2, eth2, weight 1, 00:41:14  
O>* 10.0.2.1/32 [110/20] via 192.19.2.1, eth0, weight 1, 00:40:48  
O>* 10.0.2.2/32 [110/20] via 192.19.2.1, eth0, weight 1, 00:39:51  
O>* 10.0.2.3/32 [110/20] via 192.19.2.1, eth0, weight 1, 00:29:13  
O>* 10.10.10.0/30 [110/2] via 192.19.12.2, eth1, weight 1, 00:41:03  
O 192.19.2.0/30 [110/1] is directly connected, eth0, weight 1, 00:42:06  
C>* 192.19.2.0/30 is directly connected, eth0, weight 1, 00:42:07  
L>* 192.19.2.2/32 is directly connected, eth0, weight 1, 00:42:07  
O>* 192.19.3.0/30 [110/2] via 192.19.2.1, eth0, weight 1, 00:41:21  
O 192.19.12.0/30 [110/1] is directly connected, eth1, weight 1, 00:42:06  
C>* 192.19.12.0/30 is directly connected, eth1, weight 1, 00:42:09  
L>* 192.19.12.1/32 is directly connected, eth1, weight 1, 00:42:09  
O 192.19.13.0/30 [110/1] is directly connected, eth2, weight 1, 00:42:09  
C>* 192.19.13.0/30 is directly connected, eth2, weight 1, 00:42:10  
L>* 192.19.13.1/32 is directly connected, eth2, weight 1, 00:42:10  
O>* 192.19.20.0/24 [110/2] via 192.19.12.2, eth1, weight 1, 00:40:40  
* via 192.19.13.2, eth2, weight 1, 00:40:40  
O>* 192.19.23.0/30 [110/2] via 192.19.12.2, eth1, weight 1, 00:40:35  
* via 192.19.13.2, eth2, weight 1, 00:40:35  
O>* 192.19.30.0/24 [110/2] via 192.19.12.2, eth1, weight 1, 00:40:40  
* via 192.19.13.2, eth2, weight 1, 00:40:40  
O>* 192.19.112.0/30 [110/3] via 192.19.2.1, eth0, weight 1, 00:40:49  
O>* 192.19.113.0/30 [110/3] via 192.19.2.1, eth0, weight 1, 00:40:49  
O>* 192.19.120.0/24 [110/4] via 192.19.2.1, eth0, weight 1, 00:40:02  
O>* 192.19.123.0/30 [110/4] via 192.19.2.1, eth0, weight 1, 00:40:02  
O>* 192.19.130.0/24 [110/4] via 192.19.2.1, eth0, weight 1, 00:40:02  
vyos@Router1E:~$
```

Router1E show ip route

```
vyos@Router2E: ~  
vyos@Router2E:~$ show ip route  
Codes: K - kernel route, C - connected, L - local, S - static,  
R - RIP, O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,  
T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,  
f - OpenFabric, t - Table-Direct,  
> - selected route, * - FIB route, q - queued, r - rejected, b - backup  
t - trapped, o - offload failure  
  
O>* 0.0.0.0/0 [110/10] via 192.19.12.1, eth0, weight 1, 00:56:04  
O>* 10.0.0.1/32 [110/20] via 192.19.12.1, eth0, weight 1, 00:56:04  
O>* 10.0.1.1/32 [110/20] via 192.19.12.1, eth0, weight 1, 00:56:04  
L * 10.0.1.2/32 is directly connected, lo, weight 1, 03:41:42  
C>* 10.0.1.2/32 is directly connected, lo, weight 1, 03:41:42  
O>* 10.0.1.3/32 [110/20] via 192.19.20.253, eth2, weight 1, 00:55:39  
* via 192.19.23.2, eth1, weight 1, 00:55:39  
* via 192.19.30.254, eth3, weight 1, 00:55:39  
O>* 10.0.2.1/32 [110/20] via 192.19.12.1, eth0, weight 1, 00:55:52  
O>* 10.0.2.2/32 [110/20] via 192.19.12.1, eth0, weight 1, 00:54:55  
O>* 10.0.2.3/32 [110/20] via 192.19.12.1, eth0, weight 1, 00:44:18  
O 10.10.10.0/30 [110/1] is directly connected, tun0, weight 1, 03:41:28  
C>* 10.10.10.0/30 is directly connected, tun0, weight 1, 03:41:39  
L>* 10.10.10.1/32 is directly connected, tun0, weight 1, 03:41:39  
O>* 192.19.2.0/30 [110/2] via 192.19.12.1, eth0, weight 1, 00:56:05  
O>* 192.19.3.0/30 [110/3] via 192.19.12.1, eth0, weight 1, 00:56:05  
O 192.19.12.0/30 [110/1] is directly connected, eth0, weight 1, 00:56:48  
C>* 192.19.12.0/30 is directly connected, eth0, weight 1, 00:56:50  
L>* 192.19.12.2/32 is directly connected, eth0, weight 1, 00:56:50  
O>* 192.19.13.0/30 [110/2] via 192.19.12.1, eth0, weight 1, 00:55:40  
* via 192.19.20.253, eth2, weight 1, 00:55:40  
* via 192.19.23.2, eth1, weight 1, 00:55:40  
* via 192.19.30.254, eth3, weight 1, 00:55:40  
O 192.19.20.0/24 [110/1] is directly connected, eth2, weight 1, 00:56:48  
C>* 192.19.20.0/24 is directly connected, eth2, weight 1, 00:56:50  
L>* 192.19.20.254/32 is directly connected, eth2, weight 1, 00:56:50  
O 192.19.23.0/30 [110/1] is directly connected, eth1, weight 1, 00:56:51  
C>* 192.19.23.0/30 is directly connected, eth1, weight 1, 00:56:51  
L>* 192.19.23.1/32 is directly connected, eth1, weight 1, 00:56:51  
O 192.19.30.0/24 [110/1] is directly connected, eth3, weight 1, 00:56:48  
C>* 192.19.30.0/24 is directly connected, eth3, weight 1, 00:56:48  
L>* 192.19.30.253/32 is directly connected, eth3, weight 1, 00:56:48  
L 192.19.30.253/32 is directly connected, eth3 inactive, weight 1, 03:41:40  
O>* 192.19.112.0/30 [110/4] via 192.19.12.1, eth0, weight 1, 00:55:53  
O>* 192.19.113.0/30 [110/4] via 192.19.12.1, eth0, weight 1, 00:55:53  
O>* 192.19.120.0/24 [110/5] via 192.19.12.1, eth0, weight 1, 00:55:06  
O>* 192.19.123.0/30 [110/5] via 192.19.12.1, eth0, weight 1, 00:55:06  
O>* 192.19.130.0/24 [110/5] via 192.19.12.1, eth0, weight 1, 00:55:06  
vyos@Router2E:~$
```

Router2E show ip route


```
vyos@Router3E: ~  
vyos@Router3E:~$ show ip route  
Codes: K - kernel route, C - connected, L - local, S - static,  
        R - RIP, O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,  
        T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,  
        f - OpenFabric, t - Table-Direct,  
        > - selected route, * - FIB route, q - queued, r - rejected, b - backup  
        t - trapped, o - offload failure  
  
O>* 0.0.0.0/0 [110/10] via 192.19.13.1, eth0, weight 1, 00:52:25  
O>* 10.0.0.1/32 [110/20] via 192.19.13.1, eth0, weight 1, 00:52:25  
O>* 10.0.1.1/32 [110/20] via 192.19.13.1, eth0, weight 1, 00:52:25  
O>* 10.0.1.2/32 [110/20] via 192.19.20.254, eth3, weight 1, 00:51:45  
    *                               via 192.19.23.1, eth1, weight 1, 00:51:45  
    *                               via 192.19.30.253, eth2, weight 1, 00:51:45  
L * 10.0.1.3/32 is directly connected, lo, weight 1, 03:35:49  
C>* 10.0.1.3/32 is directly connected, lo, weight 1, 03:35:49  
O>* 10.0.2.1/32 [110/20] via 192.19.13.1, eth0, weight 1, 00:51:55  
O>* 10.0.2.2/32 [110/20] via 192.19.13.1, eth0, weight 1, 00:51:06  
O>* 10.0.2.3/32 [110/20] via 192.19.13.1, eth0, weight 1, 00:40:28  
O>* 10.10.10.0/30 [110/2] via 192.19.20.254, eth3, weight 1, 00:51:46  
    *                               via 192.19.23.1, eth1, weight 1, 00:51:46  
    *                               via 192.19.30.253, eth2, weight 1, 00:51:46  
O>* 192.19.2.0/30 [110/2] via 192.19.13.1, eth0, weight 1, 00:52:26  
O>* 192.19.3.0/30 [110/3] via 192.19.13.1, eth0, weight 1, 00:52:26  
O>* 192.19.12.0/30 [110/2] via 192.19.13.1, eth0, weight 1, 00:51:46  
    *                               via 192.19.20.254, eth3, weight 1, 00:51:46  
    *                               via 192.19.23.1, eth1, weight 1, 00:51:46  
    *                               via 192.19.30.253, eth2, weight 1, 00:51:46  
O 192.19.13.0/30 [110/1] is directly connected, eth0, weight 1, 00:52:38  
C>* 192.19.13.0/30 is directly connected, eth0, weight 1, 00:52:39  
L>* 192.19.13.2/32 is directly connected, eth0, weight 1, 00:52:39  
O 192.19.20.0/24 [110/1] is directly connected, eth3, weight 1, 00:52:38  
C>* 192.19.20.0/24 is directly connected, eth3, weight 1, 00:52:38  
L>* 192.19.20.253/32 is directly connected, eth3, weight 1, 00:52:38  
O 192.19.23.0/30 [110/1] is directly connected, eth1, weight 1, 00:52:35  
C>* 192.19.23.0/30 is directly connected, eth1, weight 1, 00:52:36  
L>* 192.19.23.2/32 is directly connected, eth1, weight 1, 00:52:36  
O 192.19.30.0/24 [110/1] is directly connected, eth2, weight 1, 00:52:35  
C>* 192.19.30.0/24 is directly connected, eth2, weight 1, 00:52:38  
L>* 192.19.30.254/32 is directly connected, eth2, weight 1, 00:52:38  
O>* 192.19.112.0/30 [110/4] via 192.19.13.1, eth0, weight 1, 00:51:56  
O>* 192.19.113.0/30 [110/4] via 192.19.13.1, eth0, weight 1, 00:51:56  
O>* 192.19.120.0/24 [110/5] via 192.19.13.1, eth0, weight 1, 00:51:17  
O>* 192.19.123.0/30 [110/5] via 192.19.13.1, eth0, weight 1, 00:51:17  
O>* 192.19.130.0/24 [110/5] via 192.19.13.1, eth0, weight 1, 00:51:17  
vyos@Router3E:~$
```

Router3E show ip route

```
vyos@Router1D: ~  
vyos@Router1D:~$ show ip route  
Codes: K - kernel route, C - connected, L - local, S - static,  
R - RIP, O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,  
T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,  
f - OpenFabric, t - Table-Direct,  
> - selected route, * - FIB route, q - queued, r - rejected, b - backup  
t - trapped, o - offload failure  
  
O>* 0.0.0.0/0 [110/10] via 192.19.3.1, eth0, weight 1, 00:58:40  
O>* 10.0.0.1/32 [110/20] via 192.19.3.1, eth0, weight 1, 00:58:40  
O>* 10.0.1.1/32 [110/20] via 192.19.3.1, eth0, weight 1, 00:58:40  
O>* 10.0.1.2/32 [110/20] via 192.19.3.1, eth0, weight 1, 00:58:40  
O>* 10.0.1.3/32 [110/20] via 192.19.3.1, eth0, weight 1, 00:58:40  
L * 10.0.2.1/32 is directly connected, lo, weight 1, 11:47:33  
C>* 10.0.2.1/32 is directly connected, lo, weight 1, 11:47:33  
O>* 10.0.2.2/32 [110/20] via 192.19.112.2, eth1, weight 1, 00:57:43  
O>* 10.0.2.3/32 [110/20] via 192.19.113.2, eth2, weight 1, 00:47:06  
O>* 10.10.10.0/30 [110/2] via 192.19.113.2, eth2, weight 1, 00:47:07  
O>* 192.19.2.0/30 [110/2] via 192.19.3.1, eth0, weight 1, 00:58:41  
O 192.19.3.0/30 [110/1] is directly connected, eth0, weight 1, 00:58:49  
C>* 192.19.3.0/30 is directly connected, eth0, weight 1, 00:58:50  
L>* 192.19.3.2/32 is directly connected, eth0, weight 1, 00:58:50  
O>* 192.19.12.0/30 [110/3] via 192.19.3.1, eth0, weight 1, 00:58:41  
O>* 192.19.13.0/30 [110/3] via 192.19.3.1, eth0, weight 1, 00:58:41  
O>* 192.19.20.0/24 [110/4] via 192.19.3.1, eth0, weight 1, 00:58:41  
O>* 192.19.23.0/30 [110/4] via 192.19.3.1, eth0, weight 1, 00:58:41  
O>* 192.19.30.0/24 [110/4] via 192.19.3.1, eth0, weight 1, 00:58:41  
O 192.19.112.0/30 [110/1] is directly connected, eth1, weight 1, 00:58:46  
C>* 192.19.112.0/30 is directly connected, eth1, weight 1, 00:58:48  
L>* 192.19.112.1/32 is directly connected, eth1, weight 1, 00:58:48  
O 192.19.113.0/30 [110/1] is directly connected, eth2, weight 1, 00:58:49  
C>* 192.19.113.0/30 is directly connected, eth2, weight 1, 00:58:49  
L>* 192.19.113.1/32 is directly connected, eth2, weight 1, 00:58:49  
O>* 192.19.120.0/24 [110/2] via 192.19.112.2, eth1, weight 1, 00:47:07  
* via 192.19.113.2, eth2, weight 1, 00:47:07  
O>* 192.19.123.0/30 [110/2] via 192.19.112.2, eth1, weight 1, 00:47:07  
* via 192.19.113.2, eth2, weight 1, 00:47:07  
O>* 192.19.130.0/24 [110/2] via 192.19.112.2, eth1, weight 1, 00:47:07  
* via 192.19.113.2, eth2, weight 1, 00:47:07  
vyos@Router1D:~$
```

Router1D show ip route


```
vyos@Router2D: ~  
vyos@Router2D:~$ show ip route  
Codes: K - kernel route, C - connected, L - local, S - static,  
R - RIP, O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,  
T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,  
f - OpenFabric, t - Table-Direct,  
> - selected route, * - FIB route, q - queued, r - rejected, b - backup  
t - trapped, o - offload failure  
  
O>* 0.0.0.0/0 [110/10] via 192.19.112.1, eth0, weight 1, 01:01:44  
O>* 10.0.0.1/32 [110/20] via 192.19.112.1, eth0, weight 1, 01:01:44  
O>* 10.0.1.1/32 [110/20] via 192.19.112.1, eth0, weight 1, 01:01:44  
O>* 10.0.1.2/32 [110/20] via 192.19.112.1, eth0, weight 1, 01:01:44  
O>* 10.0.1.3/32 [110/20] via 192.19.112.1, eth0, weight 1, 01:01:44  
O>* 10.0.2.1/32 [110/20] via 192.19.112.1, eth0, weight 1, 01:01:44  
L * 10.0.2.2/32 is directly connected, lo, weight 1, 03:49:26  
C>* 10.0.2.2/32 is directly connected, lo, weight 1, 03:49:26  
O>* 10.0.2.3/32 [110/20] via 192.19.120.253, eth2, weight 1, 00:51:10  
* via 192.19.123.2, eth1, weight 1, 00:51:10  
* via 192.19.130.254, eth3, weight 1, 00:51:10  
O>* 10.10.10.0/30 [110/2] via 192.19.120.253, eth2, weight 1, 00:51:11  
* via 192.19.123.2, eth1, weight 1, 00:51:11  
* via 192.19.130.254, eth3, weight 1, 00:51:11  
O>* 192.19.2.0/30 [110/3] via 192.19.112.1, eth0, weight 1, 01:01:45  
O>* 192.19.3.0/30 [110/2] via 192.19.112.1, eth0, weight 1, 01:01:45  
O>* 192.19.12.0/30 [110/4] via 192.19.112.1, eth0, weight 1, 01:01:45  
O>* 192.19.13.0/30 [110/4] via 192.19.112.1, eth0, weight 1, 01:01:45  
O>* 192.19.20.0/24 [110/5] via 192.19.112.1, eth0, weight 1, 01:01:45  
O>* 192.19.23.0/30 [110/5] via 192.19.112.1, eth0, weight 1, 01:01:45  
O>* 192.19.30.0/24 [110/5] via 192.19.112.1, eth0, weight 1, 01:01:45  
O 192.19.112.0/30 [110/1] is directly connected, eth0, weight 1, 01:02:28  
C>* 192.19.112.0/30 is directly connected, eth0, weight 1, 01:02:31  
L>* 192.19.112.2/32 is directly connected, eth0, weight 1, 01:02:31  
O>* 192.19.113.0/30 [110/2] via 192.19.112.1, eth0, weight 1, 00:51:11  
* via 192.19.120.253, eth2, weight 1, 00:51:11  
* via 192.19.123.2, eth1, weight 1, 00:51:11  
* via 192.19.130.254, eth3, weight 1, 00:51:11  
O 192.19.120.0/24 [110/1] is directly connected, eth2, weight 1, 00:53:36  
C>* 192.19.120.0/24 is directly connected, eth2, weight 1, 01:02:32  
L>* 192.19.120.254/32 is directly connected, eth2, weight 1, 01:02:32  
O 192.19.123.0/30 [110/1] is directly connected, eth1, weight 1, 00:53:36  
C>* 192.19.123.0/30 is directly connected, eth1, weight 1, 01:02:32  
L>* 192.19.123.1/32 is directly connected, eth1, weight 1, 01:02:32  
O 192.19.130.0/24 [110/1] is directly connected, eth3, weight 1, 01:02:28  
C>* 192.19.130.0/24 is directly connected, eth3, weight 1, 01:02:28  
L>* 192.19.130.253/32 is directly connected, eth3, weight 1, 01:02:28  
L 192.19.130.253/32 is directly connected, eth3 inactive, weight 1, 03:49:25  
vyos@Router2D:~$
```

Router2D show ip route

```
vyos@Router3D:~$ show ip route
Codes: K - kernel route, C - connected, L - local, S - static,
       R - RIP, O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric, t - Table-Direct,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

O>* 0.0.0.0/0 [110/10] via 192.19.113.1, eth0, weight 1, 00:55:05
O>* 10.0.0.1/32 [110/20] via 192.19.113.1, eth0, weight 1, 00:55:05
O>* 10.0.1.1/32 [110/20] via 192.19.113.1, eth0, weight 1, 00:55:05
O>* 10.0.1.2/32 [110/20] via 192.19.113.1, eth0, weight 1, 00:55:05
O>* 10.0.1.3/32 [110/20] via 192.19.113.1, eth0, weight 1, 00:55:05
O>* 10.0.2.1/32 [110/20] via 192.19.113.1, eth0, weight 1, 00:55:05
O>* 10.0.2.2/32 [110/20] via 192.19.120.254, eth3, weight 1, 00:55:05
    *
    *   via 192.19.123.1, eth1, weight 1, 00:55:05
    *   via 192.19.130.253, eth2, weight 1, 00:55:05
L * 10.0.2.3/32 is directly connected, lo, weight 1, 00:56:08
C>* 10.0.2.3/32 is directly connected, lo, weight 1, 00:56:08
O 10.10.10.0/30 [110/1] is directly connected, tun0, weight 1, 00:55:51
C>* 10.10.10.0/30 is directly connected, tun0, weight 1, 00:56:04
L>* 10.10.10.2/32 is directly connected, tun0, weight 1, 00:56:04
O>* 192.19.2.0/30 [110/3] via 192.19.113.1, eth0, weight 1, 00:55:06
O>* 192.19.3.0/30 [110/2] via 192.19.113.1, eth0, weight 1, 00:55:06
O>* 192.19.12.0/30 [110/4] via 192.19.113.1, eth0, weight 1, 00:55:06
O>* 192.19.13.0/30 [110/4] via 192.19.113.1, eth0, weight 1, 00:55:06
O>* 192.19.20.0/24 [110/5] via 192.19.113.1, eth0, weight 1, 00:55:06
O>* 192.19.23.0/30 [110/5] via 192.19.113.1, eth0, weight 1, 00:55:06
O>* 192.19.30.0/24 [110/5] via 192.19.113.1, eth0, weight 1, 00:55:06
O>* 192.19.112.0/30 [110/2] via 192.19.113.1, eth0, weight 1, 00:55:06
    *
    *   via 192.19.120.254, eth3, weight 1, 00:55:06
    *   via 192.19.123.1, eth1, weight 1, 00:55:06
    *   via 192.19.130.253, eth2, weight 1, 00:55:06
O 192.19.113.0/30 [110/1] is directly connected, eth0, weight 1, 00:55:51
C>* 192.19.113.0/30 is directly connected, eth0, weight 1, 00:56:05
L>* 192.19.113.2/32 is directly connected, eth0, weight 1, 00:56:05
O 192.19.120.0/24 [110/1] is directly connected, eth3, weight 1, 00:55:51
C>* 192.19.120.0/24 is directly connected, eth3, weight 1, 00:56:05
L>* 192.19.120.253/32 is directly connected, eth3, weight 1, 00:56:05
O 192.19.123.0/30 [110/1] is directly connected, eth1, weight 1, 00:55:51
C>* 192.19.123.0/30 is directly connected, eth1, weight 1, 00:56:04
L>* 192.19.123.2/32 is directly connected, eth1, weight 1, 00:56:04
O 192.19.130.0/24 [110/1] is directly connected, eth2, weight 1, 00:55:51
C>* 192.19.130.0/24 is directly connected, eth2, weight 1, 00:56:05
L>* 192.19.130.254/32 is directly connected, eth2, weight 1, 00:56:05
vyos@Router3D:~$
```

Router3D show ip route

Para cada um dos servidores, mostrar ping aos outros

```
pvi@Debian01:~$ ping -c2 192.19.30.1
PING 192.19.30.1 (192.19.30.1) 56(84) bytes of data.
64 bytes from 192.19.30.1: icmp_seq=2 ttl=63 time=0.970 ms
64 bytes from 192.19.30.1: icmp_seq=2 ttl=63 time=0.948 ms

--- 192.19.30.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 0.948/0.959/0.970/0.011 ms
pvi@Debian01:~$ ping -c2 192.19.120.1
PING 192.19.120.1 (192.19.120.1) 56(84) bytes of data.
64 bytes from 192.19.120.1: icmp_seq=1 ttl=59 time=1.73 ms
64 bytes from 192.19.120.1: icmp_seq=2 ttl=59 time=2.94 ms

--- 192.19.120.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 1.729/2.332/2.935/0.603 ms
pvi@Debian01:~$ ping -c2 192.19.130.1
PING 192.19.130.1 (192.19.130.1) 56(84) bytes of data.
64 bytes from 192.19.130.1: icmp_seq=1 ttl=59 time=1.95 ms
64 bytes from 192.19.130.1: icmp_seq=2 ttl=59 time=2.69 ms

--- 192.19.130.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 1.952/2.321/2.690/0.369 ms
pvi@Debian01:~$
```

```
pvi@Debian02:~$ ping -c2 192.19.20.1
PING 192.19.20.1 (192.19.20.1) 56(84) bytes of data.
64 bytes from 192.19.20.1: icmp_seq=1 ttl=63 time=0.833 ms
64 bytes from 192.19.20.1: icmp_seq=2 ttl=63 time=0.808 ms

--- 192.19.20.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 0.808/0.820/0.833/0.012 ms
pvi@Debian02:~$ ping -c2 192.19.120.1
PING 192.19.120.1 (192.19.120.1) 56(84) bytes of data.
64 bytes from 192.19.120.1: icmp_seq=1 ttl=59 time=1.80 ms
64 bytes from 192.19.120.1: icmp_seq=2 ttl=59 time=2.57 ms

--- 192.19.120.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 1.795/2.181/2.568/0.386 ms
pvi@Debian02:~$ ping -c2 192.19.130.1
PING 192.19.130.1 (192.19.130.1) 56(84) bytes of data.
64 bytes from 192.19.130.1: icmp_seq=1 ttl=59 time=1.95 ms
64 bytes from 192.19.130.1: icmp_seq=2 ttl=59 time=2.49 ms

--- 192.19.130.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 1.951/2.220/2.490/0.269 ms
pvi@Debian02:~$
```

Debian01 Cada um dos servidores pinga os outros Debian02

```
[pvi@Alma01 ~]$ ping -c2 192.19.20.1
PING 192.19.20.1 (192.19.20.1) 56(84) bytes of data.
64 bytes from 192.19.20.1: icmp_seq=1 ttl=59 time=1.65 ms
64 bytes from 192.19.20.1: icmp_seq=2 ttl=59 time=1.68 ms

--- 192.19.20.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 1.645/1.662/1.679/0.017 ms
[pvi@Alma01 ~]$ ping -c2 192.19.30.1
PING 192.19.30.1 (192.19.30.1) 56(84) bytes of data.
64 bytes from 192.19.30.1: icmp_seq=1 ttl=59 time=1.65 ms
64 bytes from 192.19.30.1: icmp_seq=2 ttl=59 time=2.69 ms

--- 192.19.30.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 1.648/2.167/2.687/0.519 ms
[pvi@Alma01 ~]$ ping -c2 192.19.130.1
PING 192.19.130.1 (192.19.130.1) 56(84) bytes of data.
64 bytes from 192.19.130.1: icmp_seq=1 ttl=63 time=0.954 ms
64 bytes from 192.19.130.1: icmp_seq=2 ttl=63 time=0.984 ms

--- 192.19.130.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 0.954/0.969/0.984/0.015 ms
[pvi@Alma01 ~]$
```

```
[pvi@Alma02 ~]$ ping -c2 192.19.20.1
PING 192.19.20.1 (192.19.20.1) 56(84) bytes of data.
64 bytes from 192.19.20.1: icmp_seq=1 ttl=59 time=1.74 ms
64 bytes from 192.19.20.1: icmp_seq=2 ttl=59 time=2.58 ms

--- 192.19.20.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 1.737/2.156/2.575/0.419 ms
[pvi@Alma02 ~]$ ping -c2 192.19.30.1
PING 192.19.30.1 (192.19.30.1) 56(84) bytes of data.
64 bytes from 192.19.30.1: icmp_seq=1 ttl=59 time=2.03 ms
64 bytes from 192.19.30.1: icmp_seq=2 ttl=59 time=2.58 ms

--- 192.19.30.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 2.026/2.302/2.579/0.276 ms
[pvi@Alma02 ~]$ ping -c2 192.19.120.1
PING 192.19.120.1 (192.19.120.1) 56(84) bytes of data.
64 bytes from 192.19.120.1: icmp_seq=1 ttl=63 time=0.872 ms
64 bytes from 192.19.120.1: icmp_seq=2 ttl=63 time=1.18 ms

--- 192.19.120.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 0.872/1.023/1.175/0.151 ms
[pvi@Alma02 ~]$
```

Alma01 Cada um dos servidores pinga os outros Alma02

Para cada um dos servidores, ligar por ssh aos outros

```
pvi@Debian01:~$ ssh pvi@192.19.30.1
pvi@192.19.30.1's password:
Linux Debian02 6.1.0-35-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.137-1 (2025-05-07) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sun Jul 13 22:17:45 2025 from 192.19.20.1
pvi@Debian02:~$ exit
logout
Connection to 192.19.30.1 closed.
pvi@Debian01:~$
pvi@Debian01:~$ ssh pvi@192.19.120.1
pvi@192.19.120.1's password:
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Sun Jul 13 22:14:23 2025 from 192.19.130.1
[pvi@Alma01 ~]$ exit
logout
Connection to 192.19.120.1 closed.
pvi@Debian01:~$
pvi@Debian01:~$ ssh pvi@192.19.130.1
pvi@192.19.130.1's password:
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Sun Jul 13 22:12:10 2025 from 192.19.120.1
[pvi@Alma02 ~]$ exit
logout
Connection to 192.19.130.1 closed.
pvi@Debian01:~$
```

```
pvi@Debian02:~$ ssh pvi@192.19.20.1
pvi@192.19.20.1's password:
Linux Debian01 6.1.0-35-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.137-1 (2025-05-07) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Jul 19 12:25:46 2025 from 192.19.130.1
pvi@Debian01:~$ exit
logout
Connection to 192.19.20.1 closed.
pvi@Debian02:~$
pvi@Debian02:~$ ssh pvi@192.19.120.1
pvi@192.19.120.1's password:
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Sun Jul 13 22:17:20 2025 from 192.19.20.1
[pvi@Alma01 ~]$ exit
logout
Connection to 192.19.120.1 closed.
pvi@Debian02:~$
pvi@Debian02:~$ ssh pvi@192.19.130.1
pvi@192.19.130.1's password:
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Sun Jul 13 22:16:48 2025 from 192.19.20.1
[pvi@Alma02 ~]$ exit
logout
Connection to 192.19.130.1 closed.
pvi@Debian02:~$
```

Debian01 Cada um dos servidores liga por ssh aos outros Debian02

```
[pvi@Alma01 ~]$ ssh pvi@192.19.20.1
pvi@192.19.20.1's password:
Linux Debian01 6.1.0-35-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.137-1 (2025-05-07) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Jul 19 12:23:44 2025 from 192.19.120.1
pvi@Debian01:~$ exit
logout
Connection to 192.19.20.1 closed.
[pvi@Alma01 ~]$
[pvi@Alma01 ~]$ ssh pvi@192.19.30.1
pvi@192.19.30.1's password:
Linux Debian02 6.1.0-35-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.137-1 (2025-05-07) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sun Jul 13 22:13:01 2025 from 192.19.120.1
pvi@Debian02:~$ exit
logout
Connection to 192.19.30.1 closed.
[pvi@Alma01 ~]$
[pvi@Alma01 ~]$ ssh pvi@192.19.130.1
pvi@192.19.130.1's password:
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Sun Jul 13 22:07:33 2025 from 192.19.20.1
[pvi@Alma02 ~]$ exit
logout
Connection to 192.19.130.1 closed.
[pvi@Alma01 ~]$
```

```
[pvi@Alma02 ~]$ ssh pvi@192.19.20.1
pvi@192.19.20.1's password:
Linux Debian01 6.1.0-35-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.137-1 (2025-05-07) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Jul 19 12:24:15 2025 from 192.19.120.1
pvi@Debian01:~$ exit
logout
Connection to 192.19.20.1 closed.
[pvi@Alma02 ~]$
[pvi@Alma02 ~]$ ssh pvi@192.19.30.1
pvi@192.19.30.1's password:
Linux Debian02 6.1.0-35-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.137-1 (2025-05-07) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sun Jul 13 22:13:27 2025 from 192.19.120.1
pvi@Debian02:~$ exit
logout
Connection to 192.19.30.1 closed.
[pvi@Alma02 ~]$
[pvi@Alma02 ~]$ ssh pvi@192.19.120.1
pvi@192.19.120.1's password:
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Sun Jul 13 22:07:49 2025 from 192.19.20.1
[pvi@Alma01 ~]$ exit
logout
Connection to 192.19.120.1 closed.
[pvi@Alma02 ~]$
```

Alma01 Cada um dos servidores liga por ssh aos outros Alma02

B. Testes de VPN

vyos@Router2E: ~

vyos@Router2E:~\$ ip -br a
lo 127.0.0.1/8 10.0.1.2/32 fe80::200:ff:fe00:0/64 ::1/128
eth0 UP 192.19.12.2/30 fe80::a00:27ff:fe5:6d75/64
eth1 UP 192.19.23.1/30 fe80::a00:27ff:fe49:60d2/64
eth2 UP 192.19.20.254/24 192.19.20.246/24 fe80::a00:27ff:fe35:37df/64
eth3 UP 192.19.30.253/24 192.19.30.246/24 fe80::a00:27ff:fe6b:a982/64
pim6reg@NONE UNKNOWN
gre0@NONE DOWN
gretap@NONE DOWN
erspan@NONE DOWN
tun0@NONE UNKNOWN 10.10.10.1/30 fe80::f8f2:78ff:feeb:90b0/64
vyos@Router2E:~\$

vyos@Router3D: ~

vyos@Router3D:~\$ ip -br a
lo 127.0.0.1/8 10.0.2.3/32 fe80::200:ff:fe00:0/64 ::1/128
eth0 UP 192.19.113.2/30 fe80::a00:27ff:fe68:c728/64
eth1 UP 192.19.123.2/30 fe80::a00:27ff:fe5d:63af/64
eth2 UP 192.19.130.254/24 fe80::a00:27ff:fe9f:d90d/64
eth3 UP 192.19.120.253/24 fe80::a00:27ff:febf:11e0/64
pim6reg@NONE UNKNOWN
gre0@NONE DOWN
gretap@NONE DOWN
erspan@NONE DOWN
tun0@NONE UNKNOWN 10.10.10.2/30 fe80::f5ed:e0ff:febe:4e40/64
vyos@Router3D:~\$

Apresentação da Interface de rede: ip -br a

vyos@Router2E: ~

GNU nano 7.2 /etc/hosts.deny
Bloqueio por IP de Origem
sshd: 192.19.113.2
sshd: 192.19.123.2
sshd: 192.19.130.254
sshd: 192.19.130.1
sshd: 192.19.120.253
sshd: 192.19.120.1
[
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^Y Replace ^U Paste ^J Justify ^_ Go To Line

vyos@Router3D: ~

GNU nano 7.2 /etc/hosts.deny
Bloqueio por IP de Origem
sshd: 192.19.12.2
sshd: 192.19.23.1
sshd: 192.19.20.254
sshd: 192.19.20.1
sshd: 192.19.30.253
sshd: 192.19.30.1
[
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^Y Replace ^U Paste ^J Justify ^_ Go To Line

Router2E Bloqueio das outras ligações: sudo nano /etc/hosts.deny Router3D

pvi@Debian01: ~

pvi@Debian01:~\$ ssh vyos@192.19.113.2
kex_exchange_identification: read: Connection reset by peer
Connection reset by 192.19.113.2 port 22
pvi@Debian01:~\$

vyos@Router2E: ~

vyos@Router2E:~\$ ssh vyos@192.19.113.2
kex_exchange_identification: read: Connection reset by peer
Connection reset by 192.19.113.2 port 22
vyos@Router2E:~\$
vyos@Router2E:~\$ ssh vyos@192.19.123.2
kex_exchange_identification: read: Connection reset by peer
Connection reset by 192.19.123.2 port 22
vyos@Router2E:~\$
vyos@Router2E:~\$ ssh vyos@192.19.130.254
kex_exchange_identification: read: Connection reset by peer
Connection reset by 192.19.130.254 port 22
vyos@Router2E:~\$
vyos@Router2E:~\$ ssh vyos@192.19.120.253
kex_exchange_identification: read: Connection reset by peer
Connection reset by 192.19.120.253 port 22
vyos@Router2E:~\$
vyos@Router2E:~\$ ssh vyos@10.10.10.2 <----- VPN Tunnel
vyos@10.10.10.2's password:
Welcome to VyOS!

[
VyOS 2025.05.09-0019-rolling
current

* Documentation: https://docs.vyos.io/en/latest
* Project news: https://blog.vyos.io
* Bug reports: https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright

WARNING: This VyOS system is not a stable long-term support version and
is not intended for production use.
Last login: Sat Jul 19 13:21:43 2025 from 192.19.130.1
vyos@Router3D:~\$ exit
logout
Connection to 10.10.10.2 closed.
vyos@Router2E:~\$

pvi@Alma02: ~

[pvi@Alma02 ~]\$ ssh vyos@192.19.12.2
kex_exchange_identification: read: Connection reset by peer
Connection reset by 192.19.12.2 port 22
[pvi@Alma02 ~]\$

vyos@Router3D: ~

vyos@Router3D:~\$ ssh vyos@192.19.12.2
kex_exchange_identification: read: Connection reset by peer
Connection reset by 192.19.12.2 port 22
vyos@Router3D:~\$
vyos@Router3D:~\$ ssh vyos@192.19.23.1
kex_exchange_identification: read: Connection reset by peer
Connection reset by 192.19.23.1 port 22
vyos@Router3D:~\$
vyos@Router3D:~\$ ssh vyos@192.19.20.254
kex_exchange_identification: read: Connection reset by peer
Connection reset by 192.19.20.254 port 22
vyos@Router3D:~\$
vyos@Router3D:~\$ ssh vyos@192.19.30.253
kex_exchange_identification: read: Connection reset by peer
Connection reset by 192.19.30.253 port 22
vyos@Router3D:~\$
vyos@Router3D:~\$ ssh vyos@10.10.10.1 <----- VPN Tunnel
vyos@10.10.10.1's password:
Welcome to VyOS!

[
VyOS 2025.05.09-0019-rolling
current

* Documentation: https://docs.vyos.io/en/latest
* Project news: https://blog.vyos.io
* Bug reports: https://vyos.dev

You can change this banner using "set system login banner post-login" command.

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you can check individual component licenses under /usr/share/doc/*/copyright

WARNING: This VyOS system is not a stable long-term support version and
is not intended for production use.
Last login: Tue Jul 15 16:05:54 2025 from 192.19.20.1
vyos@Router2E:~\$ exit
logout
Connection to 10.10.10.1 closed.
vyos@Router3D:~\$

ssh vyos@10.10.10.2 Teste de ligação por VPN ssh vyos@10.10.10.1

pvi@Debian01: ~

```
pvi@Debian01:~$ ssh pvi@192.19.130.1
pvi@192.19.130.1's password:
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Sun Jul 20 23:25:33 2025 from 192.19.20.1
[pvi@Alma02 ~]$
[pvi@Alma02 ~]$ ssh 192.19.20.1
pvi@192.19.20.1's password:
Linux Debian01 6.1.0-35-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.137-1 (2025-05-07) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Jul 19 12:31:05 2025 from 192.19.30.1
pvi@Debian01:~$
```

vyos@Router2E: ~

```
vyos@Router2E:~$ sudo netstat -ntp | grep 10.10.10
vyos@Router2E:~$
```

vyos@Router3D: ~

```
vyos@Router3D:~$ sudo netstat -ntp | grep 10.10.10.
vyos@Router3D:~$
```

ssh do Debian1 -> Alma02 -> Debian1

Em ambos os routers, confirma-se que a ligação ssh direta entre os servidores não passa pelo túnel VPN

pvi@Alma02:~

```
pvi@Debian01:~$ ssh vyos@192.19.20.254
vyos@192.19.20.254's password:
Welcome to VyOS!
...
Last login: Sun Jul 20 12:47:18 2025 from 192.19.20.1
vyos@Router2E:~$ ssh vyos@10.10.10.2
vyos@10.10.10.2's password:
Welcome to VyOS!
...
Last login: Sun Jul 20 17:41:26 2025 from 192.19.123.1
vyos@Router3D:~$ ssh pvi@192.19.130.1
pvi@192.19.130.1's password:
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Mon Jul 14 02:47:26 2025 from 192.19.20.1
[pvi@Alma02 ~]$
```

vyos@Router3D: ~

```
vyos@Router3D:~$ sudo netstat -ntp | grep 10.10.10.
tcp        0      0 10.10.10.2:22      10.10.10.1:60122    ESTABLISHED
9341/sshd: vyos [pr
vyos@Router3D:~$
```

ssh do Debian1 -> Router2E -> Router3D -> Alma02

Confirmação de ssh estabelecido através da VPN

C. Testes do protocolo VRRP

Router2E e Router3E

```
vyos@Router2E:~$ show vrrp
Name      Interface  VRID  State      Priority  Last Transition
-----
LAN20     eth2       20    MASTER     200      12m28s
LAN30     eth3       30    MASTER     100      12m28s
vyos@Router2E:~$
```

```
vyos@Router3E:~$ show vrrp
Name      Interface  VRID  State      Priority  Last Transition
-----
LAN20     eth3       20    BACKUP     100      12m19s
LAN30     eth2       30    BACKUP     200      12m19s
vyos@Router3E:~$
```

Router2E Execução do comando show vrrp nos routers Router3E

Após desligar o cabo de eth0, eth1, eth3 no Router2E

```
vyos@Router2E:~$ show vrrp
Name      Interface  VRID  State      Priority  Last Transition
-----
LAN20     eth2       20    FAULT      200      14m45s
LAN30     eth3       30    FAULT      100      14m45s
vyos@Router2E:~$
vyos@Router2E:~$
vyos@Router2E:~$ show vrrp
Name      Interface  VRID  State      Priority  Last Transition
-----
LAN20     eth2       20    MASTER     200      13s
LAN30     eth3       30    MASTER     100      13s
vyos@Router2E:~$
vyos@Router2E:~$
vyos@Router2E:~$
```

```
pvi@Debian01:~$
vyos@Router3E:~$ show vrrp
Name      Interface  VRID  State      Priority  Last Transition
-----
LAN20     eth3       20    MASTER     100      7m57s
LAN30     eth2       30    MASTER     200      7m57s
vyos@Router3E:~$
vyos@Router3E:~$ sudo poweroff
Broadcast message from root@Router3E on pts/1 (Sun 2025-07-20 12:06:45 UTC):

The system will power off now!

vyos@Router3E:~$ Connection to 192.19.20.253 closed by remote host.
Connection to 192.19.20.253 closed.
pvi@Debian01:~$
```

Router2E Execução do comando show vrrp nos routers Router3E
É feito um segundo teste, desligando o router3E

Após desligar o cabo de eth0, eth1, eth2 no Router3E

```
pvi@Debian01:~$
vyos@Router2E:~$ show vrrp
Name      Interface  VRID  State      Priority  Last Transition
-----
LAN20     eth2       20    MASTER     200      45s
LAN30     eth3       30    MASTER     100      45s
vyos@Router2E:~$
vyos@Router2E:~$ sudo poweroff
Broadcast message from root@Router2E on pts/1 (Sun 2025-07-20 11:25:05 UTC):

The system will power off now!

vyos@Router2E:~$ Connection to 192.19.20.254 closed by remote host.
Connection to 192.19.20.254 closed.
pvi@Debian01:~$
```

```
vyos@Router3E:~$ show vrrp
Name      Interface  VRID  State      Priority  Last Transition
-----
LAN20     eth3       20    FAULT      100      8s
LAN30     eth2       30    FAULT      200      8s
vyos@Router3E:~$
vyos@Router3E:~$
vyos@Router3E:~$ show vrrp
Name      Interface  VRID  State      Priority  Last Transition
-----
LAN20     eth3       20    MASTER     100      22s
LAN30     eth2       30    MASTER     200      22s
vyos@Router3E:~$
```

Router2E Execução do comando show vrrp nos routers Router3E
É feito um segundo teste, desligando o router2E

Router2D e Router3D

vyos@Router2D: ~						vyos@Router3D: ~					
vyos@Router2D:~\$ show vrrp						vyos@Router3D:~\$ show vrrp					
Name	Interface	VRID	State	Priority	Last Transition	Name	Interface	VRID	State	Priority	Last Transition
LAN120	eth2	120	MASTER	200	33m9s	LAN120	eth3	120	BACKUP	100	29m
LAN130	eth3	130	MASTER	100	33m9s	LAN130	eth2	130	BACKUP	200	29m
vyos@Router2D:~\$						vyos@Router3D:~\$					

Router2D Execução do comando show vrrp nos routers Router3D

Após desligar o cabo de eth0, eth1, eth3 no Router2D

vyos@Router2D: ~						vyos@Router3D: ~					
vyos@Router2D:~\$ show vrrp						vyos@Router3D:~\$ show vrrp					
Name	Interface	VRID	State	Priority	Last Transition	Name	Interface	VRID	State	Priority	Last Transition
LAN120	eth2	120	FAULT	200	18s	LAN120	eth3	120	MASTER	100	22s
LAN130	eth3	130	FAULT	100	18s	LAN130	eth2	130	MASTER	200	22s
vyos@Router2D:~\$						vyos@Router3D:~\$					

Router2D Execução do comando show vrrp nos routers Router3D
Ligação ssh de Debian01 -> Router2E -> Router3D -> Alma01 -> Router2D

Após desligar o cabo de eth0, eth1, eth2 no Router3D

vyos@Router2D: ~						vyos@Router3D: ~					
vyos@Router2D:~\$ show vrrp						vyos@Router3D:~\$ show vrrp					
Name	Interface	VRID	State	Priority	Last Transition	Name	Interface	VRID	State	Priority	Last Transition
LAN120	eth2	120	MASTER	200	1m16s	LAN120	eth3	120	FAULT	100	12s
LAN130	eth3	130	MASTER	100	1m16s	LAN130	eth2	130	FAULT	200	12s
vyos@Router2D:~\$						vyos@Router3D:~\$					

Router2D Execução do comando show vrrp nos routers Router3D
Ligação ssh de Debian01 -> Router2D -> Alma01 -> Router3D

Nota1: Não foram feitos os testes extra de poweroff como para os routers Router2E e Router3E, por serem funcionalmente equivalentes a desligar os cabos das placas de rede.

Nota2: Neste exercício, seguiu-se o exemplo em aula, utilizando em ambos os routers:

```
set high-availability vrrp sync-group sync member 'LAN20'
```

```
set high-availability vrrp sync-group sync member 'LAN30'
```

e

```
set high-availability vrrp sync-group sync member 'LAN120'
```

```
set high-availability vrrp sync-group sync member 'LAN130'
```

Isto parece ter o efeito de passar ambas as redes para MASTER ou BACKUP em cada Router, apesar de terem sido definidas diferentes prioridades por rede (ver páginas 35 a 38). Pelo que entendo, isto não impacta o FailOver, mas sim o balanceamento, pois podemos ter em cada Router um MASTER numa rede e um BACKUP noutra.

Configurações

Configurações iniciais

RouterC

Teclado

```
sudo loadkeys /usr/share/keymaps/i386/qwerty/pt-latin1.kmap.gz
```

Mudança do nome

```
configure
```

```
set system host-name RouterC
```

```
commit
```

```
save
```

```
exit
```

Interfaces de Rede

```
configure
```

```
show interfaces
```

```
set interfaces ethernet eth0 address dhcp
```

```
set interfaces ethernet eth0 description 'OUTSIDE'
```

```
set interfaces ethernet eth1 address '192.19.2.1/30'
```

```
set interfaces ethernet eth1 description 'Rede2'
```

```
set interfaces ethernet eth2 address '192.19.3.1/30'
```

```
set interfaces ethernet eth2 description 'Rede3'
```

```
commit
```

```
save
```

```
exit
```

```
show interfaces
```

Configuração do serviço de terminal seguro

```
configure
```

```
set service ssh port '22'
```

```
set service ssh access-control allow user vynos
```

```
set service ssh access-control deny user root
```

```
show service ssh
```

```
commit
```

```
save
```

```
exit
```

Router1E

Teclado

```
sudo loadkeys /usr/share/keymaps/i386/qwerty/pt-latin1.kmap.gz
```

Mudança do nome

```
configure
set system host-name Router1E
commit
save
exit
```

Interfaces de Rede

```
configure
show interfaces
set interfaces ethernet eth0 address '192.19.2.2/30'
set interfaces ethernet eth0 description 'Rede2'

set interfaces ethernet eth1 address '192.19.12.1/30'
set interfaces ethernet eth1 description 'Rede12'

set interfaces ethernet eth2 address '192.19.13.1/30'
set interfaces ethernet eth2 description 'Rede13'
commit
save
exit
show interfaces
```

Configuração do serviço de terminal seguro

```
configure
set service ssh port '22'
set service ssh access-control allow user vyos
set service ssh access-control deny user root
show service ssh
commit
save
exit
```


Router1D

Teclado

```
sudo loadkeys /usr/share/keymaps/i386/qwerty/pt-latin1.kmap.gz
```

Mudança do nome

```
configure
set system host-name Router1D
commit
save
exit
```

Interfaces de Rede

```
configure
show interfaces
set interfaces ethernet eth0 address '192.19.3.2/30'
set interfaces ethernet eth0 description 'Rede3'

set interfaces ethernet eth1 address '192.19.112.1/30'
set interfaces ethernet eth1 description 'Rede112'

set interfaces ethernet eth2 address '192.19.113.1/30'
set interfaces ethernet eth2 description 'Rede113'
commit
save
exit
show interfaces
```

Configuração do serviço de terminal seguro

```
configure
set service ssh port '22'
set service ssh access-control allow user vyos
set service ssh access-control deny user root
show service ssh
commit
save
exit
```

Router2E

Teclado

```
sudo loadkeys /usr/share/keymaps/i386/qwerty/pt-latin1.kmap.gz
```

Mudança do nome

```
configure
set system host-name Router2E
commit
save
exit
```

Interfaces de Rede

```
configure
show interfaces
set interfaces ethernet eth0 address '192.19.12.2/30'
set interfaces ethernet eth0 description 'Rede12'

set interfaces ethernet eth1 address '192.19.23.1/30'
set interfaces ethernet eth1 description 'Rede23'

set interfaces ethernet eth2 address '192.19.20.254/24'
set interfaces ethernet eth2 description 'Rede20'
commit
save
exit
show interfaces
```

Configuração do serviço de terminal seguro

```
configure
set service ssh port '22'
set service ssh access-control allow user vyos
set service ssh access-control deny user root
show service ssh
commit
save
exit
```

Router3E

Teclado

```
sudo loadkeys /usr/share/keymaps/i386/qwerty/pt-latin1.kmap.gz
```

Mudança do nome

```
configure
set system host-name Router3E
commit
save
exit
```

Interfaces de Rede

```
configure
show interfaces
set interfaces ethernet eth0 address '192.19.13.2/30'
set interfaces ethernet eth0 description 'Rede13'

set interfaces ethernet eth1 address '192.19.23.2/30'
set interfaces ethernet eth1 description 'Rede23'

set interfaces ethernet eth2 address '192.19.30.254/24'
set interfaces ethernet eth2 description 'Rede30'
commit
save
exit
show interfaces
```

Configuração do serviço de terminal seguro

```
configure
set service ssh port '22'
set service ssh access-control allow user vyos
set service ssh access-control deny user root
show service ssh
commit
save
exit
```

Router2D

Teclado

```
sudo loadkeys /usr/share/keymaps/i386/qwerty/pt-latin1.kmap.gz
```

Mudança do nome

```
configure
set system host-name Router2D
commit
save
exit
```

Interfaces de Rede

```
configure
show interfaces
set interfaces ethernet eth0 address '192.19.112.2/30'
set interfaces ethernet eth0 description 'Rede112'

set interfaces ethernet eth1 address '192.19.123.1/30'
set interfaces ethernet eth1 description 'Rede123'

set interfaces ethernet eth2 address '192.19.120.254/24'
set interfaces ethernet eth2 description 'Rede120'
commit
save
exit
show interfaces
```

Configuração do serviço de terminal seguro

```
configure
set service ssh port '22'
set service ssh access-control allow user vyos
set service ssh access-control deny user root
show service ssh
commit
save
exit
```

Router3D

Teclado

```
sudo loadkeys /usr/share/keymaps/i386/qwerty/pt-latin1.kmap.gz
```

Mudança do nome

```
configure
set system host-name Router3D
commit
save
exit
```

Interfaces de Rede

```
configure
show interfaces
set interfaces ethernet eth0 address '192.19.113.2/30'
set interfaces ethernet eth0 description 'Rede113'

set interfaces ethernet eth1 address '192.19.123.2/30'
set interfaces ethernet eth1 description 'Rede123'

set interfaces ethernet eth2 address '192.19.130.254/24'
set interfaces ethernet eth2 description 'Rede130'
commit
save
exit
show interfaces
```

Configuração do serviço de terminal seguro

```
configure
set service ssh port '22'
set service ssh access-control allow user vyos
set service ssh access-control deny user root
show service ssh
commit
save
exit
```

Configuração de NAT no RouterC

Configuração de NAT com mascaramento na interface "eth0" como a interface de saída

```
configure
```

```
set nat source rule 100 outbound-interface name 'eth0'  
set nat source rule 100 source address '192.19.0.0/16'  
set nat source rule 100 translation address masquerade
```

```
## Colocar no router de saída - OSPF
```

```
## Para propagação da saída da rede através do OSPF
```

```
#set protocols ospf default-information originate always
```

```
#set protocols ospf default-information originate metric 10
```

```
#set protocols ospf default-information originate metric-type 2
```

```
commit
```

```
save
```


Configuração OSPF MultiArea

RouterC

configure

```
set interfaces loopback lo address 10.0.0.1/32
set protocols ospf parameters router-id 10.0.0.1
set protocols ospf area 0 network 192.19.2.0/30
set protocols ospf area 0 network 192.19.3.0/30
set protocols ospf log-adjacency-changes
set protocols ospf auto-cost reference-bandwidth '1000'
set protocols ospf redistribute connected metric-type 2
set protocols ospf redistribute connected route-map CONNECT
```

Colocar no router de saída

Eu, Router01, sou a saída da rede

```
set protocols ospf default-information originate always
set protocols ospf default-information originate metric 10
set protocols ospf default-information originate metric-type 2
```

```
set policy route-map CONNECT rule 10 action permit
set policy route-map CONNECT rule 10 match interface lo
```

```
commit
save
exit
show ip route
show ip ospf route
```

Router1E

configure

```
set interfaces loopback lo address 10.0.1.1/32
set protocols ospf parameters router-id 10.0.1.1
set protocols ospf area 0 network 192.19.2.0/30
set protocols ospf area 1 network 192.19.12.0/30
set protocols ospf area 1 network 192.19.13.0/30
set protocols ospf log-adjacency-changes
set protocols ospf redistribute connected metric-type 2
set protocols ospf redistribute connected route-map CONNECT
set policy route-map CONNECT rule 10 action permit
set policy route-map CONNECT rule 10 match interface lo
```

```
commit
save
exit
show ip route
show ip ospf route
```

Router2E

configure

```
set interfaces loopback lo address 10.0.1.2/32
set protocols ospf parameters router-id 10.0.1.2
set protocols ospf area 1 network 192.19.12.0/30
set protocols ospf area 1 network 192.19.23.0/30
set protocols ospf area 1 network 192.19.20.0/24
set protocols ospf log-adjacency-changes
set protocols ospf redistribute connected metric-type 2
set protocols ospf redistribute connected route-map CONNECT
set policy route-map CONNECT rule 10 action permit
set policy route-map CONNECT rule 10 match interface lo
```

```
commit
save
exit
show ip route
show ip ospf route
```

Router3E

configure

```
set interfaces loopback lo address 10.0.1.3/32
set protocols ospf parameters router-id 10.0.1.3
set protocols ospf area 1 network 192.19.13.0/30
set protocols ospf area 1 network 192.19.23.0/30
set protocols ospf area 1 network 192.19.30.0/24
set protocols ospf log-adjacency-changes
set protocols ospf redistribute connected metric-type 2
set protocols ospf redistribute connected route-map CONNECT
set policy route-map CONNECT rule 10 action permit
set policy route-map CONNECT rule 10 match interface lo
```

commit

save

exit

show ip route

show ip ospf route

Router1D

configure

```
set interfaces loopback lo address 10.0.2.1/32
set protocols ospf parameters router-id 10.0.2.1
set protocols ospf area 0 network 192.19.3.0/30
set protocols ospf area 2 network 192.19.112.0/30
set protocols ospf area 2 network 192.19.113.0/30
set protocols ospf log-adjacency-changes
set protocols ospf redistribute connected metric-type 2
set protocols ospf redistribute connected route-map CONNECT
set policy route-map CONNECT rule 10 action permit
set policy route-map CONNECT rule 10 match interface lo
```

commit

save

exit

show ip route

show ip ospf route

Router2D

configure

```
set interfaces loopback lo address 10.0.2.2/32
set protocols ospf parameters router-id 10.0.2.2
set protocols ospf area 2 network 192.19.112.0/30
set protocols ospf area 2 network 192.19.123.0/30
set protocols ospf area 2 network 192.19.120.0/24
set protocols ospf log-adjacency-changes
set protocols ospf redistribute connected metric-type 2
set protocols ospf redistribute connected route-map CONNECT
set policy route-map CONNECT rule 10 action permit
set policy route-map CONNECT rule 10 match interface lo
```

```
commit
save
exit
show ip route
show ip ospf route
```

Router3D

configure

```
set interfaces loopback lo address 10.0.2.3/32
set protocols ospf parameters router-id 10.0.2.3
set protocols ospf area 2 network 192.19.113.0/30
set protocols ospf area 2 network 192.19.123.0/30
set protocols ospf area 2 network 192.19.130.0/24
set protocols ospf log-adjacency-changes
set protocols ospf redistribute connected metric-type 2
set protocols ospf redistribute connected route-map CONNECT
set policy route-map CONNECT rule 10 action permit
set policy route-map CONNECT rule 10 match interface lo
```

```
commit
save

exit
show ip route
show ip ospf route
```

ATENÇÃO: Não é necessário para o trabalho

#Router2E - Geração do par de Chaves

```
cp -va /config/config.boot /config/config.boot.oldvpn
```

```
configure
```

```
run generate pki key-pair install ipsec-Router2E
```

```
show pki
```

```
commit
```

```
save
```

```
# RESULTADO #####
vyos@Router2E# run generate pki key-pair install ipsec-Router2E
Enter private key type: [rsa, dsa, ec] (Default: rsa)
Enter private key bits: (Default: 2048)
Note: If you plan to use the generated key on this router, do not encrypt the
private key.
Do you want to encrypt the private key with a passphrase? [y/N]
Do you want to install the public key? [Y/n] Y
Do you want to install the private key? [Y/n] Y
2 value(s) installed. Use "compare" to see the pending changes, and "commit" to
apply.
[edit]
vyos@Router2E# show pki
+key-pair ipsec-Router2E {
+  public {
+    key
MIIBIjANBgkqhkiG9w0BAQEFAA0CAQ8AMIIBCgKCAQEAwcCgAb7hFp9GYax1aJf7ebS+K/aImFJo4QyTmnjN
+iYjX5Rl31PFY3HwSB1t5MdShRe7v+sk7orNY5J1TMfFTrc8PoAYemYK28XHxwlsv+AJsPIdX0SKj/Ch0iVe
Gh8TP1ADqEhFiV6ilvL+evPa5x04Vrd1U5GX9sJba1ZvgRSws0JloW/NIp0Aj5+Y8DyFQB9SSC7V3poAUEZ
wT94zDIrPpX+8BRh0iagWKpnrEXi75wujenyVTXkhCGH+7eq28lGiTzyE6CEz/5TUCMUP12xtRjAkP9HajSO
zbiog986wRgUq2NkYmH6aZ97CpucWcQPjP3jAIp8zN7ETXftawIDAQAB
+  }
+}
```

#ATENÇÃO: Não é necessário para o trabalho

#Router3D - Geração do par de Chaves

```
cp -va /config/config.boot /config/config.boot.oldvpn
```

```
configure
```

```
run generate pki key-pair install ipsec-Router3D
```

```
show pki
```

```
commit
```

```
save
```

```
# RESULTADO #####
vyos@Router3D# run generate pki key-pair install ipsec-Router3D
Enter private key type: [rsa, dsa, ec] (Default: rsa)
Enter private key bits: (Default: 2048)
Note: If you plan to use the generated key on this router, do not encrypt the
private key.
Do you want to encrypt the private key with a passphrase? [y/N]
Do you want to install the public key? [Y/n]
Do you want to install the private key? [Y/n]
2 value(s) installed. Use "compare" to see the pending changes, and "commit" to
apply.
[edit]
vyos@Router3D# show pki
+key-pair ipsec-Router3D {
+  public {
+    key
MIIBIjANBgqhkiG9w0BAQEFAAAOCAQ8AMIIBCgKCAQEAnwwWxIJS10Py0q9/MbfyFR6PSiyuozJtpPGsLw8n
HDESA4VmH4WlTdNxv3psRdnR44fm0SazdgCZ6Pjo0dra0FoagTBHxGq5AcEGnsJwL6eG6JkIcFIXqtUyW7u9
/eN0/0+7V2YyIqe+U8ahx94KToTiBw6rLDRtaFiEPoKVVPTZvlsF5zJaVj6G92r1MTiiSjMDEW+ttpTQddWI
48hbbQduTG9a5LhwCp9IfI3c+8HaJDFow6baiM7PoUJwlytQF3Sd1vF8hPi0gReh0HBxThLHNzRhXVUsj1xG
39XGsAB3Xu4X24/bEjQiZ9efwKYwuOkUMh2yMtAKAFhTCQEfxQIDAQAB
+  }
+}
```

Configuração VPN - GRE - Site-to-Site

Router2E

configure

GRE tunnel

```
set interfaces tunnel tun0 encapsulation gre
set interfaces tunnel tun0 source-address 192.19.12.2
set interfaces tunnel tun0 remote 192.19.113.2
set interfaces tunnel tun0 address 10.10.10.1/30
```

IPsec

```
set vpn ipsec interface eth0
```

Pre-shared-secret

```
set vpn ipsec authentication psk vyos id 192.19.12.2
set vpn ipsec authentication psk vyos id 192.19.113.2
set vpn ipsec authentication psk vyos secret MYSECRETKEY
```

IKE group

```
set vpn ipsec ike-group MyIKEGroup proposal 1 dh-group '2'
set vpn ipsec ike-group MyIKEGroup proposal 1 encryption 'aes128'
set vpn ipsec ike-group MyIKEGroup proposal 1 hash 'sha1'
```

ESP group

```
set vpn ipsec esp-group MyESPGroup proposal 1 encryption 'aes128'
set vpn ipsec esp-group MyESPGroup proposal 1 hash 'sha1'
```

IPsec tunnel

```
set vpn ipsec site-to-site peer left authentication mode pre-shared-secret
set vpn ipsec site-to-site peer left authentication remote-id 192.19.113.2
```

```
set vpn ipsec site-to-site peer left ike-group MyIKEGroup
set vpn ipsec site-to-site peer left default-esp-group MyESPGroup
```

```
set vpn ipsec site-to-site peer left local-address 192.19.12.2
set vpn ipsec site-to-site peer left remote-address 192.19.113.2
```

This will match all GRE traffic to the peer

```
set vpn ipsec site-to-site peer left tunnel 1 protocol gre
```

Add GRE to the OSPF

```
set protocols ospf area 1 network 10.10.10.0/30
```

commit

save

Router3D

configure

GRE tunnel

```
set interfaces tunnel tun0 encapsulation gre
set interfaces tunnel tun0 source-address 192.19.113.2
set interfaces tunnel tun0 remote 192.19.12.2
set interfaces tunnel tun0 address 10.10.10.2/30
```

IPsec

```
set vpn ipsec interface eth0
```

Pre-shared-secret

```
set vpn ipsec authentication psk vyos id 192.19.113.2
set vpn ipsec authentication psk vyos id 192.19.12.2
set vpn ipsec authentication psk vyos secret MYSECRETKEY
```

IKE group

```
set vpn ipsec ike-group MyIKEGroup proposal 1 dh-group '2'
set vpn ipsec ike-group MyIKEGroup proposal 1 encryption 'aes128'
set vpn ipsec ike-group MyIKEGroup proposal 1 hash 'sha1'
```

ESP group

```
set vpn ipsec esp-group MyESPGroup proposal 1 encryption 'aes128'
set vpn ipsec esp-group MyESPGroup proposal 1 hash 'sha1'
```

IPsec tunnel

```
set vpn ipsec site-to-site peer right authentication mode pre-shared-secret
set vpn ipsec site-to-site peer right authentication remote-id 192.19.12.2
```

```
set vpn ipsec site-to-site peer right ike-group MyIKEGroup
set vpn ipsec site-to-site peer right default-esp-group MyESPGroup
```

```
set vpn ipsec site-to-site peer right local-address 192.19.113.2
set vpn ipsec site-to-site peer right remote-address 192.19.12.2
```

This will match all GRE traffic to the peer

```
set vpn ipsec site-to-site peer right tunnel 1 protocol gre
```

Add GRE to the OSPF

```
set protocols ospf area 2 network 10.10.10.0/30
```

commit

save

Restrições de acesso por outros caminhos Router2E

```
sudo nano /etc/hosts.deny
```

```
# Bloqueio por IP de Origem
```

```
sshd: 192.19.113.2
```

```
sshd: 192.19.123.2
```

```
sshd: 192.19.130.254
```

```
sshd: 192.19.130.1
```

```
CTRL + O -> ENTER
```

```
CTRL + X
```

```
sudo nano /etc/hosts.allow
```

```
#Permitir a ligação por IP de Origem
```

```
sshd: 10.10.10.2
```

```
CTRL + O -> ENTER
```

```
CTRL + X
```

Restrições de acesso por outros caminhos Router3D

```
sudo nano /etc/hosts.deny
```

```
# Bloqueio por IP de Origem
```

```
sshd: 192.19.12.2
```

```
sshd: 192.19.23.1
```

```
sshd: 192.19.20.254
```

```
sshd: 192.19.20.1
```

```
CTRL + O -> ENTER
```

```
CTRL + X
```

```
sudo nano /etc/hosts.allow
```

```
#Permitir a ligação por IP de Origem
```

```
sshd: 10.10.10.1
```

```
CTRL + O -> ENTER
```

```
CTRL + X
```

PRÉ-VRRP - Adição de 4ª placa de Rede

Router2E - Configuração de device, OSPF, VPN

```
configure
set interfaces ethernet eth3 address 192.19.30.253/24
set protocols ospf area 1 network 192.19.30.0/24
commit
save
```

```
sudo nano /etc/hosts.deny
sshd: 192.19.120.253
sshd: 192.19.120.1
CTRL + O -> ENTER
CTRL + X
```

Router3E - Configuração device, OSPF

```
configure
set interfaces ethernet eth3 address 192.19.20.253/24
set protocols ospf area 1 network 192.19.20.0/24
commit
save
```

Router2D - Configuração device, OSPF

```
configure
set interfaces ethernet eth3 address 192.19.130.253/24
set protocols ospf area 2 network 192.19.130.0/24
commit
save
```

Router3D - Configuração device, OSPF, VPN

```
configure
set interfaces ethernet eth3 address 192.19.120.253/24
set protocols ospf area 2 network 192.19.120.0/24
commit
save
```

```
sudo nano /etc/hosts.deny
sshd: 192.19.30.253
sshd: 192.19.30.1
CTRL + O -> ENTER
CTRL + X
```

Configuração de VRRP

Router2E

configure

```
# Rede 192.19.20.0/24 (ligada ao Debian01)
set high-availability vrrp group LAN20 vrid '20'
set high-availability vrrp group LAN20 interface 'eth2'
set high-availability vrrp group LAN20 hello-source-address '192.19.20.254'
set high-availability vrrp group LAN20 peer-address '192.19.20.253'
set high-availability vrrp group LAN20 address '192.19.20.246/24'
set high-availability vrrp group LAN20 priority '200'
delete high-availability vrrp group LAN20 no-preempt
set high-availability vrrp group LAN20 preempt 1
set high-availability vrrp group LAN20 preempt-delay 3
set high-availability vrrp group LAN20 authentication type plaintext-password
set high-availability vrrp group LAN20 authentication password Passw0rd
```

```
# Rede 192.19.30.0/24 (ligada ao Debian02)
set high-availability vrrp group LAN30 vrid '30'
set high-availability vrrp group LAN30 interface 'eth3'
set high-availability vrrp group LAN30 hello-source-address '192.19.30.253'
set high-availability vrrp group LAN30 peer-address '192.19.30.254'
set high-availability vrrp group LAN30 address '192.19.30.246/24'
set high-availability vrrp group LAN30 priority '100'
delete high-availability vrrp group LAN30 no-preempt
set high-availability vrrp group LAN30 preempt 1
set high-availability vrrp group LAN30 preempt-delay 3
set high-availability vrrp group LAN30 authentication type plaintext-password
set high-availability vrrp group LAN30 authentication password Passw0rd
```

Grupo de sincronismo

```
set high-availability vrrp sync-group sync member 'LAN20'
set high-availability vrrp sync-group sync member 'LAN30'
```

commit

save

exit

Router3E

configure

```
# Rede 192.19.30.0/24 (ligada ao Debian02)
set high-availability vrrp group LAN30 vrid '30'
set high-availability vrrp group LAN30 interface 'eth2'
set high-availability vrrp group LAN30 hello-source-address '192.19.30.254'
set high-availability vrrp group LAN30 peer-address '192.19.30.253'
set high-availability vrrp group LAN30 address '192.19.30.246/24'
set high-availability vrrp group LAN30 priority '200'
delete high-availability vrrp group LAN30 no-preempt
set high-availability vrrp group LAN30 preempt 1
set high-availability vrrp group LAN30 preempt-delay 3
set high-availability vrrp group LAN30 authentication type plaintext-password
set high-availability vrrp group LAN30 authentication password Passw0rd

# Rede 192.19.20.0/24 (ligada ao Debian01)
set high-availability vrrp group LAN20 vrid '20'
set high-availability vrrp group LAN20 interface 'eth3'
set high-availability vrrp group LAN20 hello-source-address '192.19.20.253'
set high-availability vrrp group LAN20 peer-address '192.19.20.254'
set high-availability vrrp group LAN20 address '192.19.20.246/24'
set high-availability vrrp group LAN20 priority '100'
delete high-availability vrrp group LAN20 no-preempt
set high-availability vrrp group LAN20 preempt 1
set high-availability vrrp group LAN20 preempt-delay 3
set high-availability vrrp group LAN20 authentication type plaintext-password
set high-availability vrrp group LAN20 authentication password Passw0rd

# Grupo de sincronismo
set high-availability vrrp sync-group sync member 'LAN20'
set high-availability vrrp sync-group sync member 'LAN30'

commit
save
exit
```

Router2D

configure

```
# Rede 192.19.120.0/24 (ligada ao Alma01)
set high-availability vrrp group LAN120 vrid '120'
set high-availability vrrp group LAN120 hello-source-address '192.19.120.254'
set high-availability vrrp group LAN120 peer-address '192.19.120.253'
set high-availability vrrp group LAN120 interface 'eth2'
set high-availability vrrp group LAN120 address '192.19.120.246/24'
set high-availability vrrp group LAN120 priority '200'
delete high-availability vrrp group LAN120 no-preempt
set high-availability vrrp group LAN120 preempt 1
set high-availability vrrp group LAN120 preempt-delay 3
set high-availability vrrp group LAN120 authentication type plaintext-password
set high-availability vrrp group LAN120 authentication password Passw0rd

# Rede 192.19.130.0/24 (ligada ao Alma02)
set high-availability vrrp group LAN130 vrid '130'
set high-availability vrrp group LAN130 interface 'eth3'
set high-availability vrrp group LAN130 hello-source-address '192.19.130.253'
set high-availability vrrp group LAN130 peer-address '192.19.130.254'
set high-availability vrrp group LAN130 address '192.19.130.246/24'
set high-availability vrrp group LAN130 priority '100'
delete high-availability vrrp group LAN130 no-preempt
set high-availability vrrp group LAN130 preempt 1
set high-availability vrrp group LAN130 preempt-delay 3
set high-availability vrrp group LAN130 authentication type plaintext-password
set high-availability vrrp group LAN130 authentication password Passw0rd

# Grupo de sincronismo
set high-availability vrrp sync-group sync member 'LAN120'
set high-availability vrrp sync-group sync member 'LAN130'

commit
save
exit
```

Router3D

configure

```
# Rede 192.19.130.0/24 (ligada ao Alma/Rocky02)
set high-availability vrrp group LAN130 vrid '130'
set high-availability vrrp group LAN130 interface 'eth2'
set high-availability vrrp group LAN130 hello-source-address '192.19.130.254'
set high-availability vrrp group LAN130 peer-address '192.19.130.253'
set high-availability vrrp group LAN130 address '192.19.130.246/24'
set high-availability vrrp group LAN130 priority '200'
delete high-availability vrrp group LAN130 no-preempt
set high-availability vrrp group LAN130 preempt 1
set high-availability vrrp group LAN130 preempt-delay 3
set high-availability vrrp group LAN130 authentication type plaintext-password
set high-availability vrrp group LAN130 authentication password Passw0rd

# Rede 192.19.120.0/24 (ligada ao Alma/Rocky01)
set high-availability vrrp group LAN120 vrid '120'
set high-availability vrrp group LAN120 interface 'eth3'
set high-availability vrrp group LAN120 hello-source-address '192.19.120.253'
set high-availability vrrp group LAN120 peer-address '192.19.120.254'
set high-availability vrrp group LAN120 address '192.19.120.246/24'
set high-availability vrrp group LAN120 priority '100'
delete high-availability vrrp group LAN120 no-preempt
set high-availability vrrp group LAN120 preempt 1
set high-availability vrrp group LAN120 preempt-delay 3
set high-availability vrrp group LAN120 authentication type plaintext-password
set high-availability vrrp group LAN120 authentication password Passw0rd

# Grupo de sincronismo
set high-availability vrrp sync-group sync member 'LAN120'
set high-availability vrrp sync-group sync member 'LAN130'

commit
save
exit
```

Scripts para abertura e fecho com guarda de estado das VMs

StartVMs 5104-ProjetoFinal.BAT

```
@echo off
REM Indicar os npmes das VMs EXACTAMENTE como aparecem no VBox
set VMs=5104-VyOS-RouterC 5104-VyOS-Router1E 5104-VyOS-Router2E 5104-VyOS-Router3E
5104-VyOS-Router1D 5104-VyOS-Router2D 5104-VyOS-Router3D 5104-Debian01 5104-Debian02
5104-Alma01 5104-Alma02

REM Tempo em segundos de pausa entre arranques
set PAUSA=10

REM Caminho completo para VBoxManage.exe
set VBOXM="C:\Program Files\Oracle\VirtualBox\VBoxManage.exe"

for %%i in (%VMs%) do (
    echo A arrancar %%i...
    %VBOXM% startvm "%%i" --type gui
REM    %VBOXM% startvm "%%i" --type headless
    timeout /t %PAUSA% /nobreak
)
echo Todos os pedidos de arranque foram enviados!
pause
```

SaveStateVMs 5104-ProjetoFinal.BAT

```
@echo off
REM Indicar os npmes das VMs EXACTAMENTE como aparecem no VBox
set VMs=5104-Debian01 5104-Debian02 5104-Alma01 5104-Alma02 5104-VyOS-Router3E
5104-VyOS-Router2E 5104-VyOS-Router1E 5104-VyOS-Router3D 5104-VyOS-Router2D
5104-VyOS-Router1D 5104-VyOS-RouterC

REM Tempo em segundos de pausa entre arranques
set PAUSA=10

REM Caminho completo para VBoxManage.exe
set VBOXM="C:\Program Files\Oracle\VirtualBox\VBoxManage.exe"

for %%i in (%VMs%) do (
    echo A guardar o estado de %%i...
    %VBOXM% controlvm "%%i" savestate
    timeout /t %PAUSA% /nobreak
)
echo Todos os pedidos de guardar estado foram enviados!
pause
```